EXECUTIVE SUMMARY

Ayurveda is a comprehensive scientific system of medicine evolved in India. Initially it was developed through ancient wisdom, clinical experiences and experimentation in scientific manner. At present, Ayurveda is growing in the paradigm of contemporary scientific, technological and medical parameters.

The term 'Ayurveda' meaning 'the knowledge of life' comprises of two Sanskrit words *viz.* '*Āyu*' meaning 'Life' and '*Veda*' meaning 'Knowledge' or 'Science'. The earliest concepts of medical science are found in the ancient wisdom called *Veda*, which were believed to be composed between 5000 - 1000 BC. The classical texts of Ayurveda containing exclusive information on health and disease came around 1000 B.C. onwards, when fundamental and applied principles of Ayurveda were codified in a systematic manner. The basic tenets of Ayurveda are elaborated in two great medical compendia *viz. Caraka Saṃhitā* and *Suśruta Saṃhitā*. Since its advent, Ayurveda passed through different political and socio cultural phases and stood test of time, got enriched by the contribution of different practitioners and scholars and catered the health needs of contemporary societies. Present form of Ayurveda is the outcome of continued scientific inputs that have gone into the evolution of its principles, theories and guidelines of healthy living and disease management.

At present, Ayurveda has become integral part of national health delivery system of India. Ayurveda institutions for research and education, hospitals and dispensaries were established in different parts of the country. The government had initiated measures to regulate the system. In 1971, Central Council of Indian Medicine (CCIM) was set up as statutory body to regulate education and practice. For undertaking integrated and coordinated research, Composite Drug Research Scheme (CDRS) was initiated in 1964. In 1970, Central Council for Research in Indian Medicine & Homoeopathy (CCRIMH) was established for giving focused attention towards research. A separate Department of Indian Systems of Medicine and Homoeopathy (ISM&H) under the Ministry of Health & Family Welfare, Government of India was created in 1995 for developing Ayurveda, Yoga & Naturopathy, Unani, Siddha and Homoeopathy systems. This department was renamed as Department of Ayurveda, Yoga & Naturopathy, Unani, Siddha and Homoeopathy (AYUSH) in 2003. Research activities in Ayurveda have increased manifold with establishment of separate and autonomous research council *viz.* Central Council for Research in Ayurvedic Sciences (CCRAS).

Education, clinical practice, manufacturing and sale of Ayurvedic medicines are regulated in the country through Acts, Rules and Regulations. Infrastructure development in private and public sectors has improved the outreach of Ayurveda to the community in a commendable way. Global resurgence of demand for natural medicine has filliped R&D activities in

Ayurveda at national and international levels with inter-disciplinary collaboration and linkages.

Government of India has enunciated "National Policy on Indian Systems of Medicine & Homoeopathy - 2002". The policy *inter alia* includes the strategy for comprehensive development of Ayurveda in contemporary perspective.

The Department of AYUSH has supported several international conferences/seminars/ workshops/trade fairs in collaboration with Indian Missions, Universities, Associations and other agencies promoting traditional systems of medicine overseas. AYUSH experts are frequently deputed as resource persons to such events and also for facilitating Continuing Medical Education (CME) and training programs on AYUSH systems in foreign universities. Chairs on Ayurveda are being set up in the Universities in Germany and South Africa. Under the Department's International Cooperation scheme the department of AYUSH has set up an AYUSH information cell in the premises of the Indian cultural centre in Malaysia with the support of Indian High Commission in Kuala Lumpur, Malaysia. India is a prominent member in the Inter-governmental Committee (IGC) on Traditional Knowledge, Genetic Resources and Folklore.

Good health is a basic prerequisite to acquire materialistic, social and spiritual upliftment of individual. Therefore Ayurveda lays great emphasis on preservation and promotion of health and prevention of disease. This is achieved through life style management rather than drug or any other external intervention. According to Ayurveda, the living human body is a composition of the body matrix, *Indriya* (sensory and motor organs), *Manas* (mind) and *Ātmā* (soul). Ayurveda advocates a holistic approach while understanding healthy and diseased states of the human body, and their management. In this respect the human body is considered a microcosm representing the Universe, a macrocosm. The universe as well as the human body is made up of '*Pañcamahābhūta*', the five basic elements *viz.* '*Pṛthvī*', the basic earthy/gross element, '*Jala*', the basic aqueous element, '*Agni*', the basic thermal element, '*Vāyu*', the basic gaseous element and '*Ākāśa*', the basic ethereal element. Similarly the Moon, Sun and Wind in the Universe are represented in the human body respectively by three '*Doṣa*' (regulatory and functional entities of the body) *viz.* '*Kapha*', '*Pitta*' and '*Vāyu*'. These '*Doṣa*' are regulatory and functional entities of the body. Any function in the body is grossly attributed to one of the three '*Doṣa*'.

The structural entities in the body are classified as '*Dhātu*'. They are seven in number and are responsible for maintaining the body in a compact and composed state. The seven '*Dhātu*' are i. '*Rasa*' (nutrient fluids), ii. '*Rakta*' (blood), iii. '*Māṃsa*' (muscle tissue), iv.

'*Meda*' (adipose tissue), v. '*Asthi*' (bone tissue), vi. '*Majjā*' (bone marrow) and vii. '*Śukra*' (reproductive tissue).

The structural and functional elements of the body are accompanied by excretory entities, classified as '*Mala*'. '*Mala*' are grossly divided into three *viz.* '*Mūtra*', the urine, '*Purīṣa*', the faeces and '*Sveda*', the sweat. The excretions from eyes, ear etc. are also categorized as '*Mala*'.

Apart from '*Doṣa*', '*Dhātu*' and 'Mala' the living body harbors an independent mechanism of digestion and metabolism. The digestion and metabolism is function of '*Agni*', which is a composite mechanism in various parts of the body. Ayurveda defines health as a state of equilibrium of '*Doṣa*', '*Dhātu*', '*Mala*' and '*Agni*'. Loss of this equilibrium invariably results into manifestation of disease.

Every individual has a unique '*Prakṛti*' (psychosomatic constitution) which is responsible for the structural and functional attributes of body and mind, health or disease patterns of an individual. The basic constitution of a person is determined at the time of conception. Ayurvedic concept of examination of '*Prakṛti*' is a key determinant for prescribing individualized diet, lifestyle or treatment regimen.

The disease management in Ayurveda is considered under two major heads 1. *Nidāna* (diagnosis) and 2. *Cikitsā* (treatment). Ayurvedic method of diagnosis involves *Rogī Parīkṣā* (clinical examination) as well as the *Roga Parīkṣā* (diagnosis of disease). The diagnosis is principally based on evaluating the status of *'Doṣa', 'Dhātu', 'Mala'* and *'Agni'*, whereas treatment modules are designed principally with an object of restoring their equilibrium. The Ayurvedic treatment methods can be grossly divided into three methods *'Saṃśodhana'* (bio-cleansing therapy), *'Saṃśamana'* (palliative therapy) and *'Nidāna Parivarjana'* (avoidance of causative factors). *'Saṃśodhana'* is practiced through one or more therapies from *'Panchakarma'*, whereas *'Saṃśamana'* is achieved through *'Āhāra'* (food), *'Vihāra'* (lifestyle) and *'Dravya'* (drug).

Ayurveda believes that no substance in the Universe is devoid of therapeutic potential, provided it is used judiciously. Plants, Minerals, Metals and Animal products serve as a basic source of Ayurvedic drugs. Drugs prepared from plants are known as '*Kāṣṭhauṣadhi*' whereas mineral and metal drugs are known as '*Rasauṣadhi*'. Animal products are not frequently used as independent drugs; however they are used as a medium or vehicle in preparation or administration of plant and mineral/metal drugs. The regimens are customized

by the practitioner depending upon the specific requirements in accordance with the basic '*Prakṛti*', and health status of 'the individual.

Ayurvedic pharmaceutical science can be broadly considered under two major heads 1. '*Dravya Guṇa*' (Ayurvedic pharmacology) and 2. '*Bhaiṣajya Kalpanā*' (Ayurvedic pharmaceutics). Plant drugs are used in different dosage forms in accordance with subject specific requirement. The basic dosage forms of plant drugs are known as '*Pañcavidha Kaṣāya Kalpanā*' (five dosage forms). They are '*Svarasa*' (expressed juice), '*Kalka*' (paste), '*Kvātha*' (decoction), '*Phāṇṭa*' (hot infusion) and '*Hima*' (cold infusion). Many more dosage forms such as '*Cūrṇa*' (powder), '*Vaṭī*' (pill), *Āsava- Ariṣṭa* (medicated fermented preparations), '*Lehya*' (linctus), aqueous extracts etc. derived with modern technology are also in practice. The methods of preparation of these dosage forms are fairly simple, because of which the Ayurvedic practitioners prefer to prepare the required drugs on their own. However preparation of '*Rasauṣadhi*' (mineral and metal drugs) is a complicated procedure. Minerals and metals are generally known to be potentially harmful to the human body if not processed properly. These minerals and metals are subjected to complex and meticulous processing to make them therapeutically useful and safe to the body in prescribed doses. The final product of mineral/metal drugs made with incineration generally known as '*Bhasma*' (calcined material) and others as '*Rasauṣadhi*'.

Ayurvedic drugs are being manufactured on large scale by approximately 8000 Ayurvedic drug manufacturing units spread all over the country. According to Drugs and Cosmetics Act 1940, Ayurvedic drugs are classified into two categories - i. Classical or generic drug formulations manufactured exclusively in accordance with the formulae described in the specified authoritative books of Ayurveda and ii. Patent and Proprietary drug formulations developed by the manufacturer. The demand for medicinal plants has increased manifold due to upsurge in domestic Ayurvedic drug industry and global interest in herbal products. The Government of India has set up National Medicinal Plants Board with the object to promote cultivation and propagation of medicinal plants and also to ensure sustained supply of quality plant material to Ayurvedic drug industry.

Manufacturing, sale and distribution of Ayurvedic drugs are regulated through Drugs and Cosmetic Act 1940 and Drugs and Cosmetics Rules 1945. The Government of India has set up the Ayurvedic Pharmacopoeia Committee (APC) to prescribe standards of single drugs and compound formulations mentioned in Ayurveda for the use of manufacturers. Moreover Good Manufacturing Practices (GMP) under Schedule 'T' of the Drugs and Cosmetics Act 1940, have also been notified to ensure the quality of ASU (Ayurveda, Siddha, Unani) medicines. Reference quality standards of drugs are published in Ayurvedic Pharmacopeia to

Chapter 2

FUNDAMENTAL CONCEPTS

2.1 Basic principles

2.1.1 *Pañcamahābhūta* (five basic elements) - The Universe according to Ayurveda is composed of five basic elements called *Pañcamahābhūta viz. ākāśa* (basic ethereal element), *vāyu* (basic gaseous element), *agni* (basic thermal element), *jala* (basic aqueous element) and *pṛthvī* (basic earthy/ gross element) and so is the human body. There is a fundamental similarity between universe and man. A harmonious interaction between the microcosm (human being) and the macrocosm (universe) is the basis of health.

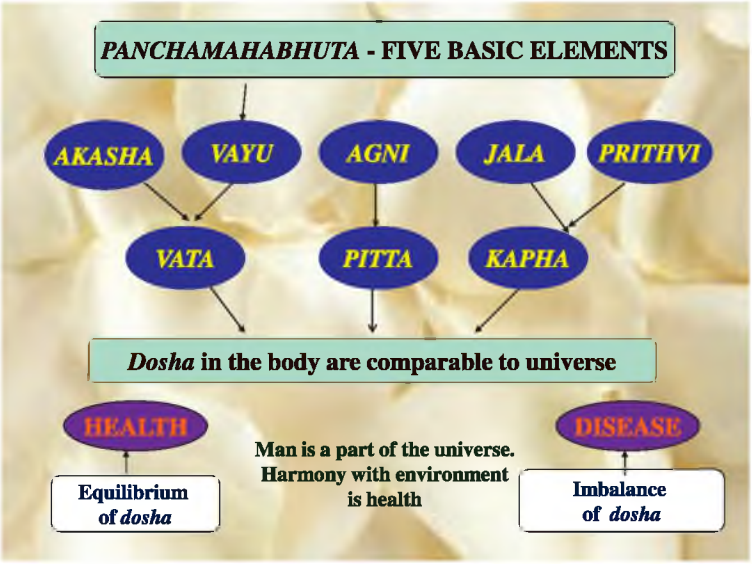

Fig. 6 - Similarity between universe and human being

2.1.2 Health and disease - Optimal health conceived in Ayurveda is a perfect harmony of body, mind and soul. Health or *"svāsthya"* is a state of equilibrium of the *doṣa* (regulatory physiological entities), *agni* (digestive and metabolic factors), function of *dhātu* (structural entities), *mala* (excretory entities) along with proper functioning of *jñānendriya* (sense organs), *manas* (cheerful mind) and *ātmā* (soul). Any disturbance in this

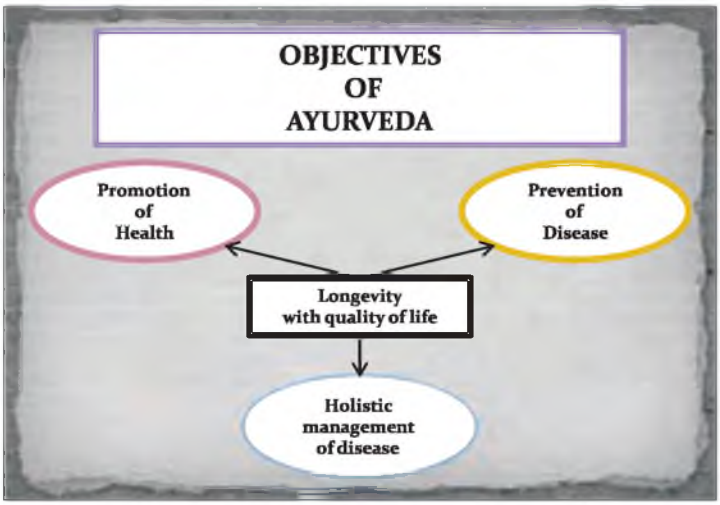

Fig. 7 – Objectives of Ayurveda

equilibrium due to internal or external factor leads to diseases. Ayurveda emphasizes that *Prakṛti* (psychosomatic constitution), which is specific to every individual, is responsible for the health or disease pattern in the individual. Human mind has *triguṇa* (three attributes) i.e. *sattva* (pure state of mind), *rajas* (mind with passion/desire/ attachment) and *tama* (inert/ignorant mind), which interact with the biological components, *vāta, pitta* & *kapha* and determine the psycho-somatic constitution of an individual. Ayurvedic approach to examination of psychosomatic constitution is important in assessing individual's health for planning preventive measures for selecting diet, medicine or treatment regimen.

2.1.3 *Tridoṣa* (Three Regulatory Physiological Entities)

The doctrine of *Pañcamahābhūta* is the origin and basis of the three regulatory physiological entities i.e. *vāta*, *pitta* and *kapha*. *Vāta* is derived from the basic element *vāyu* and *ākāśa*, *pitta* from *agni*, *kapha* from *jala* and *pṛthvī*.

i. *Vāta* - The important function of *vāta* is to impart movement, generation and conduction of impulses, transportation of biological materials and elimination of waste products. In the normal condition, *vāta* performs entire neurological functions of the body. It sustains the entire machinery of the body. *Vāta* is responsible for functioning of five sensory organs *(Pañca jñānendriya)* and motor functions also. It is a regulator of psychosomatic functions of the living body. When *vāta* is vitiated or its equilibrium is disturbed, it brings about various psycho-somatic disturbances. All the basic emotions like worry, anxiety, fear, grief, anger etc. are governed by *vāta*. Usually all functions of nervous system at central and peripheral level is correlated with *vāta*.

ii. *Pitta*- *Pitta* represents transformation. It governs digestion, absorption, assimilation, nutrition, metabolism, body temperature, skin coloration, luster of the eyes, intelligence, and understanding. Psychologically, pitta arouses anger, hate, and jealousy. Morbidity of *pitta* may lead to insomnia, abnormality of body temperature, yellow discoloration of eyes, impaired digestion/metabolism etc. Usually digestive juices, enzymes and hormones come under this entity.

iii. *Kapha*- *Kapha* is one of the primary constituent of the living body. It exhibits psychological phenomena as the exhibition of the courage, forbearance, zest, virility, knowledge, understanding etc. Similarly, the physical function of *kapha* is responsible for the physical strength, built, stability of structures, cooling, adhesion, lubrication,

maintenance of the smooth working of the joints etc. It is also expressed in tendencies toward calmness, forgiveness and love.

These *tridoṣa* are described as the main cause for health and disease. The health is maintained if they remain in balanced state and if they are deranged in any manner, they vitiate structural and excretory entities to cause *vikāra/roga* (disease).

2.1.4 *Saptadhātu* (Seven Structural Entities)

The structural elements in the body are classified as *dhātu*. They are seven in numbers and are responsible for maintaining the body in a compact and composed state. They are *rasa* (nutritional fluid), *rakta* (blood), *maṃsa* (muscle tissue), *medas* (adipose tissue), *asthi* (bone tissue), *majjā* (bone marrow) and *śukra* (reproductive elements). According to Ayurveda, *āhāra rasa* (nutrient fluid) nourish these tissues. Apart from this, there are six *upadhātu* (supportive bye products of *dhātu*), namely *stanya* (breast milk), *ārtava* (menstrual fluid), *kaṇḍarā* (tendons), *śirā* (blood vessels), *vasā* (fat), *tvak* (skin), *snāyu* (ligament).

2.1.5 *Ojas* (Essence of dhātu)

Ayurvedic texts have vividly described the factor of immunity in terms of *vyādhikṣamatva*, which is considered as the natural or acquired biological defense of an individual against diseases. This power is attributed to the presence of a biological factor called *ojas*, the essence of the structural entities (*dhātu*), which literally means vigor. Ayurveda also describes a number of methods to promote *ojas* and *vyādhikṣamatva*.

2.1.6 *Mala* (Excretory Entities)

Mūtra (urine), *purīṣa* (faeces) and *sveda* (sweat) are the three gross excretory entities. The exudates eliminated from eye, nose, mouth, ears and reproductive organs and other structural entities etc. are considered as subtle excretory entities. Proper elimination of these excretory entities is also required to maintain health. Their hypo, hyper and irregular state results in diseases.

2.1.7 *Srotas* (Micro and Macro Channels)

The concept of *srotas* occupies an important position in the development of conceptual frame work of Ayurveda. According to Ayurveda whole body is made up of *srotas* (micro and macro

channels), which transport all types of materials in the body. For normal functioning of the body, it is essential that these channels, both micro and macro remain intact.

i. *Prāṇavaha srotas* - Channels of respiration/respiratory system
ii. *Udakavaha srotas* - Channels for regulation and transportation of fluids
iii. *Annavaha srotas* - Digestive tract
iv. *Rasavaha srotas* - Channels in which nutrient fluid is formed and transported
v. *Raktavaha srotas* - Channels through which blood is formed and transported
vi. *Maṃsavaha srotas* - Channels in which muscle tissue is formed and transported
vii. *Medovaha srotas* - Channels in which adipose tissue is formed and transported
viii. *Asthivaha srotas* - Channels in which bone tissue is formed and transported
ix. *Majjāvaha srotas* - Channels in which bone marrow is formed and transported
x. *Śukravaha srotas* - Channels in which reproductive tissue is formed and transported
xi. *Mūtravaha srotas* - Channels in which urine is formed and excreted
xii. *Svedavaha srotas* - Channels in which sweat is formed and transported
xiii. *Purīṣavaha srotas* - Channels in which faeces is formed and excreted

Therefore, emphasis has been given in Ayurveda to prevent *srotoduṣṭi* (vitiation of *srotas*).

2.1.8 *Agni* (Digestive and metabolic factors)

The digestive and metabolic energy that is responsible for transformation of food to nutrients is called *agni*. It is responsible for digestion and metabolism in the body. In other words *agni* signifies life process responsible for entire digestion, endocrine and metabolic activities. It has been classified into the following 13 types:

1. *Jaṭharāgni* (digestive factors located in digestive tract)
2. Five types of *Bhūtāgni* (metabolic factors located in *pañcamahābhūta*)
3. Seven types *Dhātvagni* (metabolic factors located at *dhātu*)

Jaṭharāgni is the most important *agni* and supports the other type of *agni*. Derangement of *agni viz. mandāgni* (down regulation), *tikṣṇāgni* (hyper-active), *viṣamāgni* (deranged state) lead to diseases and *samāgni* (balanced *agni*) maintains the health of an individual.

2.1.9 *Manas* (Mind)

Mind is a principal sense organ which controls remaining five senses. The mind is called *atīndriya* (beyond the perception of senses). The sense-organs receive the stimulus from the external world and are perceived through mind and passed to the *ātmā* (soul).The chief function of *manas* is assimilation and discrimination. It has been stated that it enables the *buddhi* (intellect) to discriminate good and bad, right and wrong. It has normal functions like to think, to determine or decide, to express, to conceal, to recall, to concentrate, to memorize, to control emotion, to meditate etc. The emotional factors like lust, anger, greed, delusion and hallucination or confusion, malice, remorse and anxiety, fear, exhilaration are all responsible for psychosomatic disorders. The above description of mind clearly indicates that it is a controlling component of the body and it is largely influenced by external socio-cultural factors. The body can be prevented from the stress by controlling the mind. Patañjali in his *Yoga Sutra* has clearly propounded yoga system for the control of mind and maintenance of positive health.

2.1.10 *Prakṛti* (Psychosomatic Constitution)

The structural and functional variations are the fundamental characteristics of human being. No two individuals are exactly alike either in their morphological, physiological or behavioral dimensions. The individual differ in their genetic makeup and also in morphological and psychological aspects including their endocrine activity and metabolic efficiency. Ayurveda has given maximum attention regarding the clinical significance of individual personality and also recognized the dichotomy between mind and body and classified psychological personality in terms of *sāttvika*, *rājasika* and *tāmasika* where as somatic constitution is determined by relative predominance of *vāta*, *pitta* and *kapha* in an individual. Ayurvedic concept of *Prakṛti parīkṣā* is to know body- mind constitution to predict the susceptibility of disease. *Prakṛti* is considered while prescribing diet, medicine or treatment regimen to an individual.

Characteristics of *vāta* predominance *prakṛti* - *Vāta* type person are thin, tall, disproportionate, under-developed in general physique, chest is flat and depressed, veins are prominent and will have markedly projected ends of bone. *Vāta* type individual looks emaciated with rough dark, pale, dusty complexion and dry cracked rough skin. Psychologically these persons are characterized by short memory and low will power.

Vāta type individuals are mentally unstable due to the predominance of *vāta* activity. They have least power of reasoning and they are of irritable temperament. They are comparatively non-religious and coward. The life span of *vāta* type individual is comparatively low. Due to predominance of rajas traits, *vāta* type individuals are more susceptible to psychosomatic disorders. A comprehensive regimen of life has been advocated for the prevention of diseases for the vāta type individuals in almost all the classical texts of Ayurveda.

Characteristics of *pitta* predominance *prakṛti*- *Pitta* dominant individuals are medium in strength, stature and body built. They have fair and coppery complexion, smooth, very soft & wrinkled skin and thin, silky and brownish hair. The psychological traits of such individuals are sharp, bright, intelligent and short tempered. If we trace the origin of *pitta* personality from *triguṇa* point of view, *sattva* predominates in this constitution. Individual of this type of personality is more prone to blood borne skin and diseases. Therefore, such individuals have been advised to live in a cool place and use diet which is antagonistic to *agni mahābhūta.*

Characteristics of *kapha* predominance *prakṛti* - Physically *kapha* type people are strong with heavy & proportionate body built and fair complexion. They are biologically strong; therefore are more virile and have good progeny. From psychological point of view such individuals are mentally stable with maximum capacity for retention. In Ayurvedic texts *kapha* type of personality is considered to be the ideal type of personality because they take the balanced approach to life. They have maximum capacity to withstand stress. Since, *kapha* type people are biologically strong; they are less prone to develop psychosomatic disorders.

2.1.11 Clinical diagnosis

The diagnosis in Ayurveda is based on two-fold approach *viz.* (1) examination of the patient i.e., *rogī parīkṣā* and (2) diagnosis of the disease *i.e. roga parīkṣā*.The *rogi parīkṣā* is essentially concerned with ascertaining the psychosomatic constitution and status of health & vitality of the individual. This is achieved through ten fold examinations (*daśavidha parīkṣā*), eight fold examination (*aṣṭasthāna parīkṣā*) three fold examination (*trividha parīkṣā*) of the patient. For the proper treatment of diseases, it is mandatory to understand the exact nature of the disease with reference to *doṣa*, *dhātu*, *mala* and *agni*. The diagnosis of the disease is also done with the help of *ṣaṭkriyākāla* (six stages of pathogenesis), *nidāna pañcaka* (five fold approaches of diagnosis).

Chapter 3

LIFESTYLE MANAGEMENT

3.1 Food

The maintenance of health and prevention from diseases can be achieved through food and lifestyle specific to individual needs and in line with the seasons and cycles of nature. Food and habits which are conducive to the body are known as *pathya*. Food is referred as '*mahābhaiṣajya*' (the best medicine) by Ayurvedic classics. Ayurveda describes a large number of food and drinks, their method of preparation and the code and discipline of taking the food. The food has been given a godly status and is considered a subject of worship. The food is essentially said to have the five basic elements. Their appropriate use may help in balancing the similar elemental components of the body.

Bad eating habits are the basic cause of indigestion which is root cause for all diseases. Ayurvedic texts enlist bad eating habits such as:

- Unwholesome and non- congenial food
- Incompatible food combinations
- Consumption of food before the digestion of previously eaten food or when not hungry
- Overeating
- Too much water or no water at all during a meal
- Eating hurriedly
- Eating while emotionally upset
- Eating at the wrong time of the day
- Eating too much heavy or too light food
- Improperly preserved and stale foods
- Use of too spicy, sour, salty food

3.1.1 *Ṣaḍrasa* (six tastes of food items) - The concept of *ṣaḍrasa* is a central point in Ayurvedic cuisine. These six tastes *viz*. sweet, sour, salty, pungent, bitter and astringent

should be present in balanced proportions. Each taste has an influence on *doṣa* and hence contributes to the health or disease.

3.1.2 Eight factors of diet and dietetics: Ayurveda offers some basic dietary guidelines that include choosing appropriate quantity of food, combinations of food, cooking methods, storage, eating atmosphere, hygiene and eating etiquettes. Following eight factors related to consumption of food need to be considered:

1. Nature (*prakṛti*) – Not all the food articles are suitable for everybody. In fact the suitability of the food articles is dependent on the psychosomatic constitution of the individual. Hence it is necessary that one should always consider the original nature of the food article in the light of his psychosomatic constitution while consuming the food. In diseased state the status of the *doṣa* and effect of the food on them need to be considered.

2. Process of preparation (*saṃskāra*) - The method of processing like frying, roasting etc have a definite effect on the nature of prepared food. The processed food will have different characteristics than the original food. Appropriate processing technique can make the food suitable for the consumer in accordance with the requirements.

3. Compatibility (*saṃyoga*) – Food articles when used in combinations may either be useful or harmful. Combination of food material having opposite properties may prove harmful to the consumer. Illustrations of such harmful combinations are given in the Ayurvedic texts (*Viruddhāhāra*). The consumer should always avoid such harmful food combinations.

4. Quantity (*raśi*) – The quantity of food as a whole or item wise need to be considered. Light food also if consumed in excess is likely to be harmful. Heavy foods should always be consumed in small quantity. In this reference the stomach is considered to be divided into four parts. Out of these two parts have to be filled with solids and one part with liquids. The remaining one part should always be left empty.

5. Habitat (*deśa*) - Habitat is geographic region. Specific food is either suitable or not suitable according to variations in the climatic condition of the area.

6. Time (*kāla*) - Eating schedules need to be observed every time one takes food. Consumption of food at odd hours is detrimental to the health. Seasonal variations also need to be considered while selecting food materials.

7. Rules of eating (*upayoga saṃsthā*) - There are certain dietetic rules, which need to be followed by one and all. This ensures proper digestion and metabolism.

8. Consumer (*upayoktā*): The consumer should observe all disciplines of dietetics. Quality, quantity, processing, combinations of the food and consideration of digestive capacity, health status, age, season are important factors to be considered.

Observation of above stated guidelines plays a significant role in promotion of health and prevention of disease.

3.1.3 Diet and Mind - Considering the effect of food on mind, following 3 classifications have been made:

1. ***Sāttvika* food-** Ideal diet containing vegetarian, non-oily, non-spicy food articles which are congenial to the mind.
2. ***Rājasika* food-**Too spicy, hot, sour, salty that excites the mind.
3. ***Tāmasika* food-**Too oily, tasteless, putrefied, stale and heavy food that leads to lethargic mind.

Along with a balanced diet, incorporating other healthy habits into a daily routine can prevent disease at its root level.

3.2 Lifestyle (*Vihāra*)

3.2.1 *Svasthavṛtta* (lifestyle advocacy for maintenance of health) - *Svasthavṛtta* is a healthy life style, prescribed for the maintenance of health of an individual. Health preventive measures include diet, practices and regimen during day (*dinacaryā*), night (*rātricarya*), different seasons (*ṛtucaryā*) and code of conduct (*sadvṛtta*). Seasonal and appropriate use of *pañcakarma* and *rasāyana* are also important in the prevention of the diseases.

3.2.2 *Dinacaryā* (daily regimen)

Researchers in medical chrono-biology have long discovered that our body has many inbuilt rhythms or cycles. Most of our body functions follow a daily cycle. Our weight fluctuates during the day and is the maximum in the evening. Our body temperature is highest in the evening and most of our hormones have their well defined periods of high and low secretions during the day and the most obvious of all is the sleep – wake cycle. Therefore intrinsically the emphasis of Ayurveda is on natural cycles. *Dinacaryā* meaning the daily regimen refers to

a healthy and sustainable pattern of lifestyle. In order to be optimally healthy, one should tune the body to the nature′s master cycle which in turn regulates the various other rhythms. To achieve this, Ayurveda prescribes a specific routine in general and also on the basis of psychosomatic constitution of an individual. This includes waking up early in the morning, excretion of bio-wastes, taking care of oral hygiene (cleaning of teeth and tongue, gargling), exercise, massage, bath, clothing, sleep, eye care, nasal therapy etc. The various aspects of this daily routine in general are:

Waking up- Since our biological clock is tuned to the rising and setting of the sun, it is advisable to wake up before the sunrise in perfect synchronization to the natural clock. An ideal time to wake up is *brahmamuhūrta* (1-2 hours before sunrise).

Natural Urges- Dawn is the best time to eliminate the body's physical waste. Drinking one or two glass of warm water in winter season and normal water in other seasons before sunrise helps in proper elimination of *mala* and renders multiple health benefits.

Oral Hygiene- Brushing of teeth and cleaning the gums are advised early in the morning and after each meal by using the twigs/thin stems of different plants like *Neem* (*Azadirachta indica*), *Khadira* (*Acacia catechu*) *etc*. Scrapping the tongue using tongue cleaners made up of metals or plants are advised. The gargling with water, decoctions and medicated oils keeps gum, mouth, teeth and throat healthy.

Exercise- Exercise increases the body's stamina and resistance to disease by facilitating the immune system, clearing all channels, promoting circulation & waste disposal and destroying fat. Exercises may be in the form of yoga or walking etc. Depending on age and body type, *kapha* type can go for heavy exercises, *pitta* type should do it in moderation and *vāta* type should perform light exercises. Exercise should not be done during illness and just after meal.

Massage- The gentle oil massage everyday makes the skin supple; controls *vāta* by reducing its cold, dry, light, rough and erratic qualities; enhances blood circulation; encourages elimination of metabolic wastes and relaxes the body.

Bathing- Warm bath is advisable for the body and cold water for the head.

Clothing- Clothing should always be clean, light and made of natural fibers as cotton, wool or silk. The use of natural perfumes in moderation promotes pleasant feeling.

Sleep- Night is the natural time to sleep and day sleep is contraindicated except for the very young, old, very weak and those intoxicated, diseased, exhausted or traumatized and who could not have sufficient sleep during night. The person with disturbed sleep should massage the feet with oil before going to bed.

Brahmacarya **-** In the broad sense it means control of the senses or *indriya* especially sexual life. More specifically, it refers to celibacy or chastity. Like all traditional spi*ritu*al traditions, Ayurveda advocates restraining from indulging in sensual gratification. The more broad definition of *brahmacarya* also includes conduct that leads to the realization of the self. The conservation of energy that comes from practicing celibacy is converted into *ojas*. Many people mistakenly believe that practicing *brahmacarya* means suppression of the natural urge but on the contrary anything that is suppressed will eventually lead to disorder.

Eye care- Washing the eyes every morning with clean and cold water or decoction of *triphalā* is very useful. Use of a medicated collerium (*sauvīrāñjana*) every day is also recommended for the healthy eyes.

Nasya (nasal instillations) - Regular inhalation of the *Aṇu taila*, the oil prepared by boiling 26 different medicinal plants in gingelly oil and goat's milk prevents all ailments of the eyes, hair, nose and the ears.

3.2.3 *Adhāraṇīya Vega* (non suppressible natural urges) - There are thirteen natural urges, suppression of which leads to many diseases as given below:

1. Suppression of urge of urination may lead to difficulty in passing urine, urinary stone, atony of bladder and inflammation of urinary tract.
2. Suppression of bowel movement may lead to pain in abdomen, indigestion, gas in abdomen, headache.
3. Suppression of flow of flatus may lead to pain in abdomen, indigestion, heart diseases, constipation or diarrhoea.
4. Suppression of flow of semen may produce pain in testis and difficulty in intercourse.
5. Suppression of urge for vomiting may lead to different types of diseases like urticaria, giddiness, anaemia, hyperacidity, skin diseases and fever.
6. Suppression of sneezing may produce rhinitis and chronic cold, headache, sinusitis and diseases of respiratory system.

7. Suppression of eructation may lead to hiccough, pain in chest, cough and loss of appetite.
8. Suppression of yawning may lead to diseases of the eyes, throat, ear and nose.
9. Suppression of hunger may lead to indigestion, nutritional disorders and debility.
10. Suppression of thirst may lead to nutritional disorders and debility.
11. Suppression of tears may lead to mental disorders, pain in chest, giddiness and digestive disorders.
12. Suppression of respiration after exertion may cause suffocation, respiratory disorders, heart diseases and even death.
13. Suppression of sleep may cause diseases like insomnia, mental disorders, digestive disorders and diseases of sense organs.

3.2.4 *Dhāraṇīya vega* (suppressible natural urges) - Apart from 13 non-suppressible natural urges mentioned above, Ayurveda describes some suppressible natural urges. One should control the urges of greed, envy, hatred, jealousy, lust etc. and gain control over the worldly pleasures.

3.2.5 *Ṛtucaryā* (Seasonal Regimen)

According to Ayurveda the *doṣa* and the *ṛtu* (seasons) are interlinked. Health is affected by the nature of the climate as outer environment influences the body. For example, when the air is damp, cold and wet it increases these qualities in the body leading to aggravation of *kapha*, which has similar qualities. Hence there is an increase in mucous, catarrh and colds in winter. There are various environmental factors like temperature, humidity, wind, rain, clouds and atmospheric pressure and sunlight etc. that affect individual's health. *Ṛtucaryā* is the observance of diet and regimen according to the seasonal changes. In Ayurveda, a year is divided into two *kāla* (cycles) or periods based on the apparent position of the sun in the north and southern directions. They are:

1. *Uttarāyaṇa* - Northern solstice
2. *Dakṣiṇāyaṇa* - Southern solstice

Uttarāyaṇa is also called *ādanakāla* or the taking away period. The sun and wind are powerful during this period. The energy of the body is diminished. Due to the heat, air becomes hot and drains the cooling effect of earth. Due to this, people get dehydrated and weakened and the atmosphere becomes hot and dry.

Dakṣiṇāyaṇa is also called *visargakāla* or the giving away period. The moon becomes powerful during this period. The earth cools down due to cold winds and rain. People regain their strength and nourishment that was lost in the *ādanakāla*.

Each year consists of six *ṛtu* or seasons. Each *ṛtu* comprises of two months and three such *ṛtu* constitute one *kāla*. Hence, *ādānakala* and *visargakāla* each comprises of six months and three *ṛtu*. The six *ṛtu* and their characteristics are summarized below in the table:

***Kāla* (Semester)**	***Ṛtu* (Season)**	**Month**	**Characteristics**
Ādāna (*Uttarāyaṇa*)	*Śiśira*	Mid January to mid March	Winter and spring
	Vasanta	Mid March to mid May	Spring
	Grīṣma	Mid May to mid July	Summer
Visarga (*Dakṣiṇāyaṇa*)	*Varṣā*	Mid July to mid September	Rainy Season
	Śarad	Mid September to mid November	Autumn
	Hemanta	Mid November to mid January	Dewy season

This classification is based on Indian climate. This may differ from area to area where mainly three seasons namely summer, rainy and winter occur.

Seasonal diet and lifestyle

***Hemanta Ṛtu* (dewy season)** ***Śiśira Ṛtu* (winter and spring)** In both these seasons the *agni* becomes more powerful. *Vāyu* is accentuated and needs to be pacified by comparatively heavy diet.	
Diet	**Lifestyle**
• Intake of food with sweet, sour and salt taste helps pacifying the *vāta* • Wine prepared from *jaggery* (molasses) can be taken • Wheat/gram flour products, milk products, sugarcane products and corn/edible oils can be taken as a part of food	• *Abhyaṅga*- Massage with oil followed by steam bath • Dry body massage (*Udavartana*) • Exercise. • Clothing-leather, silk and wool. • Exposure to sunlight and heat to keep oneself warm.

• Carrots, tomatoes, figs, dates, cane sugar, nuts to be taken • Warm water should be taken as it aids digestion • Bitter, astringent and pungent food must be avoided	• Right time for having sexual pleasure

***Vasanta Ṛtu* (spring)**	
During *vasanta ṛtu*, increased *kapha* liquefied by the heat of sun causes diminished *agni*	
• Take easily digestible food. Barley, honey, roasted meat, mango juice. • Beverages such as *āsava* - *ariṣṭa* (Medicated fermented preparations), *sīdhu* (fermented sugarcane juice), honey mixed with water and water boiled with extracts of sandal wood. • Avoid food that is heavy to digest, cold, sour, sweet and fatty. Such food increase *kapha* causing *doṣa* imbalance and hence genesis of disease.	• Vigorous physical exercise. • Dry massage. • Nasal instillation can be done after massage and bath with camphor, sandalwood and saffron • Avoid sleep during the day

***Grīṣma Ṛtu* (summer)**	
Grīṣma ṛtu is the season of dehydration, exhaustion, lack of energy and lethargy. During summer, there occurs decrease of *kapha* due to the heat of the sun and dryness in the air.	
• Sweet, light, fatty and liquid food can be taken. • Boiled rice with meat, corn flour, buttermilk (yoghurt) can be taken in food. • Drink cold water from clay pot. • Syrup prepared with grapes, sugarcane, resin, dates, *kāśmarya* (*Gmelina arboria*) and *paruṣaka* (*Phoenix pusilla*) fruits all in equal quantity cold with cardamom powder. • Fresh juices and juicy fruits, salads, buttermilk can be taken in abundance.	• Anoint body with sandal wood paste and take bath with cold water. • Stay in cool places. • Wear loose and light cotton dresses to keep the body cool. • Use perfumes made from substances that possess cold properties • Sleep during day is permitted as nights are short. • Too much exertion and sunshine should be avoided

***Varṣā Ṛtu* (rainy season)**	
The *agni* weakens further and gets vitiated by *vāta*. Lack of sunshine as well as a cloudy atmosphere is non-congenial to health.	
• Easily digestible food such as pulses, juice, soups, old grains and *mastu* (watery portion of yoghurt) can be taken in food. • Ginger, black pepper and lemon juice may be taken to reinforce appetite. • Leafy vegetables should be taken sparingly. • Foods should be hot and light with ghee, curd and honey. • Should be careful about water contaminated with rain.	• *Pañcakarma* can be done. • Perfumes can be used. • Avoid sleep in during daytime. • Avoid exertion and too much exposure to sunlight.
***Śarad Ṛtu* (Autumn Season)**	
The damp and hot atmosphere aggravates *pitta*. Sudden exposure to sunlight after the rains and cool atmosphere increases *pitta*.	
• Ghee processed with bitter herbs can be taken. • Intake of bitter, astringent and sweet taste food items is useful. • Take easily digestible food like rice, green gram, Indian gooseberry, honey and sugar. • Avoid heavy food, curd, oil, strong liquors.	• *Udavartana* with *candana*. • Bath with warm water. • Pearls give soothing effect from aggravated *pitta*. • Avoid direct exposure to breeze, liquor and sleep during day

***Ṛtusandhi* and its significance-** *Sandhi* means conjunction. *Ṛtusandhi* is the period of the last 7 days of the preceding season and the first seven days of forthcoming season. During *ṛtusandhi*, the diet and regimen that is being followed should be given up gradually and that of the subsequent season *may* be taken up similarly.

3.2.6 *Sadvṛtta* (code of conduct)

The foundation of Ayurveda lies in its basic principle that accepts the holistic nature and the integrity of body, mind, and soul. A healthy mind is as important as a healthy body. This ethical regimen contains principles of right conduct that are applicable to all; irrespective of caste, creed and religion. Practice of code of conduct helps to keep the body and mind in a balanced condition. Emphasis is given on righteous and truthful behavior, possession of high moral, abstinence from greed, hatred, passion and envy in this code of conduct. Discipline in (1) *āhāra* (food), (2) *nidrā* (sleep) (3) *brahmacarya* (celibacy) called *trayopastambha* are basis of fruitful life and good health.

Thus *dinacaryā*, *ṛtucaryā and sadvṛtta* form the basis of healthy living.

Note to the readers: The readers may obtain more information from the books, journals and websites mentioned in the concerned section of the document such as:

1. Agnivesa. Caraka Saṃhitā English translation by Dash Bhagwan, Sharma RK. Varanasi, India: Chaukhambha Publications, 2nd Edition, 2000.
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4. Vagabhata. Astanga Samgraha Commentary by Dasai R. 2nd Edition, Nagpur, India: Baidyanath Ayurveda Bhawan; 1981.
5. Suśruta. Suśruta Saṃhitā. 5th Edition. New Delhi,India: Motilal Banarasidas; 1994.